REAL-TIME NETWORK INTRUSION DETECTION

SAGE

Safe Adaptive In-Kernel Neural Inference via eBPF

```
$ sudo ./sage --attach eth0
```

The Problem

- Modern DDoS attacks exceed hundreds of Gbps
- User-space ML pipelines: too slow — milliseconds per packet
- Need: microsecond-level packet classification at line rate

KERNEL SPACE

~1 µs

XDP / eBPF hook

vs

USER SPACE

~5 ms

Context switch + ML

eBPF: Power with Constraints

No floating point

512B stack limit

1M instruction limit

No dynamic allocation

No signed division

"How do you fit a neural network in here?"

Three Contributions

01 **Formal Overflow Safety**
Closed-form theorem for int64 accumulator safety bounds

02 **Design Space Exploration**
48 configurations, Pareto-front analysis for accuracy vs resources

03 **End-to-End System**
4-class IDS deployed in XDP, 806 BPF instructions, 1.8KB model

01 — Overflow Safety

THEOREM

$$b \leq \lfloor (63 - \log_2 C_k) / 2 \rfloor$$

where C_k = max fan-in at layer k, b = scale factor bits

⚠ THEORY SAYS

b=16 is UNSAFE

b_{max} = 14 for safe guarantee

✓ EMPIRICAL REALITY

5,590× headroom

Actual max ≪ theoretical max

Theoretical max 2^{63}

Actual max

02 — 48-Config Exploration

Search space: hidden_size × layers × scale_bits

b=16 is the critical threshold — accuracy saturates beyond this point

Finding: h=8 already saturates at 99.6% accuracy

Pareto Optimal

Config	Accuracy	Instructions	Memory	
h=8	99.6%	754	656B	smallest
h=16	99.6%	754	1.8KB	sweet spot
h=32	99.6%	806	5.6KB	
h=64	99.6%	806	19.5KB	

30× less memory, same accuracy

03 — Four-Stage Tail-Call Pipeline

Packet ingress → **Entry** 309 instrs → **Layer 0** 109 instrs → **Layer 1** 152 instrs → **Layer 2** 236 instrs → **PASS / DROP** XDP action

Per-CPU	Zero	806
Scratch Map	Lock Contention	Total Instructions

Results

99.6%

Classification Accuracy

4	200K	0.98	754	1.8KB
Classes	Samples	Macro-F1	BPF Instrs	Model Memory

Benign · DDoS · PortScan · BruteForce

vs. Prior Work

System	Classes	Accuracy	Memory
DT-eBPF	2	99%	32KB+
NN-int8	2	~95%	~2KB
NN-eBPF	2	99.2%	5KB
SAGE (ours)	**4**	**99.6%**	**1.8KB**

More classes · Higher accuracy · Less memory

Key Takeaways

- ✓ **b=16** is the architecture-invariant critical threshold for quantization scale
- ✓ Theoretical overflow bound is **5,590× conservative** — empirical safety far exceeds formal guarantees
- ✓ **h=16 is Pareto-optimal**: maximum accuracy with minimal resource footprint (1.8KB, 754 instrs)

Thank You

github.com/usernameisveryok/sage-ebpf-ml

```
$ echo "Questions?"
```